

WEEKLY REPORT-1

Completed Tasks:

1. Consolidated the initial internship orientation carried out at the start of the engagement and clarified that the system must support basin-scale hydro-climatic visualization for Himalayan analysis in a locally usable web environment.
2. Reviewed the planned project assets, including ERA5-Land tables, future CMIP6 projections, basin and glacier boundary layers, PMTiles-based map resources, and the provision for adding model and user-uploaded datasets later.
3. Inspected the repository structure and separated the solution into backend API, frontend dashboard, database, map assets, outcomes, and packaging modules so the development workflow remained well organized.
4. Defined the baseline scientific workflow for the application: dataset selection, year-range selection, variable loading, spatial visualization, elevation filtering, and time-series based regional interpretation.
5. Prepared the initial system architecture in which the backend performs indexed parquet reads, filtering, and aggregation, while the frontend handles map rendering, controls, legends, and graph interaction.

Reporting No: 1 Week No: 1

Date From: 25/05/2026 Date To: 30/05/2026

Student ID: 23DCS076

Internship Title: WebApp for Visualization and Analysis of Himalayan Hydro-Cryospheric Datasets

Figure:



Name	Date modified	Type	Size
.git	09-07-2026 02:10	File folder	
.github	31-03-2026 21:25	File folder	
backend	26-05-2026 06:26	File folder	
Database	09-07-2026 02:04	File folder	
frontend	09-07-2026 02:10	File folder	
Glacier_shp	26-05-2026 00:04	File folder	
Himalaya_shape	29-03-2026 08:15	File folder	
Map_handle	02-04-2026 22:25	File folder	
Outcomes	24-05-2026 22:03	File folder	
Reports	09-07-2026 02:53	File folder	
SPHY_Model_data_prep	19-06-2026 00:00	File folder	
.dockerignore	29-03-2026 13:56	DOCKERIGNORE F...	1 KB
.gitattributes	29-03-2026 07:06	Git Attributes Sour...	1 KB
.gitignore	31-03-2026 23:49	Git Ignore Source ...	1 KB
backend.zip	01-04-2026 00:58	Compressed (zipp...	47 KB
CONFIGURATION.md	25-02-2026 20:49	Markdown Source...	8 KB
Dockerfile	29-03-2026 13:56	File	1 KB
HF_DEPLOYMENT.md	29-03-2026 13:57	Markdown Source...	2 KB
pack_offline.ps1	03-04-2026 02:00	Windows PowerS...	8 KB
PROJECT_SUMMARY.txt	25-02-2026 20:49	Text Document	20 KB
QUICKSTART.md	25-02-2026 20:49	Markdown Source...	4 KB

Figure 1. repository structure view.

Plans for next week:

1. Prepare reusable CSV-to-parquet conversion logic for the initial ERA5 data batches and verify that the exported schema remains consistent across yearly files.
2. Start implementing FastAPI endpoints for datasets, dates, variables, statistics, and filtered map responses.
3. Design the first backend indexing flow so incremental data additions can be handled without reworking the whole application structure.

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References:

1. Project discussions and scientific requirements shared during the ISRO SAC internship initiation.
2. Repository structure and setup notes from README.md, QUICKSTART.md, and CONFIGURATION.md.
3. Initial ERA5-Land and CMIP6 export notes reviewed for dataset planning.

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